Research Methodology

Complete Practical Exam Study Guide for Unseen Research Paper Analysis

Beginner-friendly, exam-focused, syllabus-aligned notes with examples, answer templates, viva preparation, and final revision toolkit.

DSE Research Methodology - Practical Exam Preparation

Prepared from the uploaded syllabus and unit materials covering Research Fundamentals, Problem Identification, Data Analytics, Presenting Research Outcomes, Publication Process, and Research Ethics.

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1. How to Use This Guide Tonight

A practical plan for a student seeing the subject for the first time.

Your practical exam is likely to test whether you can read an unseen research paper and identify the important research-methodology parts: problem, objectives, methodology, data, findings, limitations, ethics, and future scope. This guide is written for that exact purpose.

Best study order for last-minute preparation

1. First read Chapter 9 and Chapter 10 so you know how to attack an unseen paper quickly.
2. Then read Unit 1 and Unit 2 because most viva questions come from basic definitions and problem identification.
3. Read Unit 3 for dataset, preprocessing, statistics, and hypothesis testing terms.
4. Read Unit 4 and Unit 5 for research paper structure, publication, peer review, plagiarism, and journal selection.
5. Revise Unit 6 ethics because it is short and easy to score.
6. Finally, memorize the answer templates and the final revision sheet.

What your examiner is actually checking

- **Can you locate information in a paper?** For example, finding objectives in the introduction or methodology in the methods section.
- **Can you use correct research vocabulary?** Words like research gap, population, sample, methodology, limitation, hypothesis, and data collection method must be used correctly.
- **Can you justify your answer?** Do not only say "survey method". Say why: "The authors collected responses from participants using a structured questionnaire."
- **Can you speak clearly in viva?** A short direct answer followed by one example is usually better than a long memorized answer.

One sentence survival rule
In the exam, every answer should follow this pattern: **Identify + Evidence from the paper + Explanation + Limitation/importance.** Example: "The study uses a quantitative survey methodology because the authors collected numerical responses through a questionnaire and analyzed them statistically."

2. Complete Syllabus Map and Exam Priorities

Syllabus Unit	What it includes	How it appears in practical/viva
Unit 1 - Research Fundamentals	Meaning, significance, requirements, characteristics, types of research, research process, induction, deduction, tools, qualities of researcher.	Definitions, differences, examples, viva basics.
Unit 2 - Problem Identification	Choosing a problem area, sources of research articles, literature review, evaluating research problem, techniques, methodologies, state of the art.	Research problem, gap, objectives, questions, literature review.
Unit 3 - Data Analytics	Exploring and organizing datasets, preprocessing, interpreting data, descriptive statistics, association, inferential statistics, population parameters, hypothesis testing.	Dataset, preprocessing, data analysis technique, statistics terms.
Unit 4 - Presenting Research Outcomes	Elements of research paper, problem, methods and data, data analysis and interpretation, weaknesses, presenting output, conclusions.	Paper structure, findings, limitations, conclusion, report writing.
Unit 5 - Publication	Journal submission and review, peer review, single blind and double blind, societies, scientometric analysis, citation index, plagiarism checker.	Peer review, plagiarism, journal selection, citation metrics.
Unit 6 - Research Ethics	Protection from harm, informed participation, privacy, conflict of interest, honesty, code of ethics, IP rights, fraud and misconduct.	Ethical issues in given paper and viva answers.

High scoring practical areas

- Research problem, objectives, research gap, methodology, sample/population, sampling technique, data collection, tools/instruments, data analysis, findings, limitations, future scope, ethics.
- These terms must be identified from the paper, not guessed from the title only.
- Always support your answer with words from the paper: "questionnaire", "participants", "dataset", "experiment", "interview", "statistical analysis", "model accuracy", etc.

3. Unit 1 - Research Fundamentals

3.1 Meaning of Research

Research is a systematic and logical investigation used to discover new knowledge, verify existing knowledge, solve a problem, or answer a question. In simple language, research is not just searching on Google; it is a planned process of asking a meaningful question, collecting evidence, analyzing that evidence, and presenting a justified conclusion.

Aspect	Explanation	Example
Systematic	Research follows ordered steps instead of random guessing.	Define problem -> review literature -> collect data -> analyze -> conclude.
Logical	Each step should make sense and be connected to the objective.	If the objective is to study student stress, data should come from students, not unrelated groups.
Empirical	Good research should be based on observation, data, documents, experiments, or evidence.	Using survey responses or experimental results.
Replicable	Another researcher should be able to repeat the process and check the result.	Mentioning dataset, sample size, tools, and method clearly.

> **How to write in exam**
> Research is a systematic, logical, and evidence-based investigation carried out to discover new facts, verify existing theories, solve problems, or answer research questions. It involves problem identification, literature review, data collection, analysis, interpretation, and reporting.

3.2 Significance of Research

- **Creates knowledge:** It adds new facts, theories, models, or explanations.
- **Solves practical problems:** Applied research helps solve real-world issues in health, education, business, computer science, governance, and society.
- **Supports decisions:** Research provides evidence for policy, product design, teaching methods, and technical choices.
- **Improves existing systems:** Research can compare current methods and propose better techniques.
- **Prevents assumptions:** Instead of relying on opinion, research uses data and logic.

3.3 Requirements and Characteristics of Good Research

Characteristic	Meaning	Exam example
Clear problem	The study should investigate a specific issue.	"Effect of online classes on attention span" is clearer than "online classes".
Objective	The researcher should avoid personal bias.	Do not collect only positive reviews for a method.
Reliable	The method should produce stable results if repeated under similar conditions.	Same questionnaire should measure the same concept consistently.
Valid	The method should measure what it claims to measure.	A programming test measures coding skill better than asking students whether they feel good at coding.
Ethical	Participants and data must be handled responsibly.	Informed consent and privacy protection.
Feasible	The study should be possible with available time, data, skills, and resources.	A semester project should not require nationwide data collection unless data is already available.
Original	It should add some value instead of simply copying earlier work.	New dataset, new comparison, new context, or improved method.

3.4 Types of Research

Type	Meaning	Simple example
Basic / Pure research	Research done mainly to increase knowledge, not necessarily to solve an immediate practical problem.	Studying the theoretical limits of machine learning algorithms.
Applied research	Research done to solve a real practical problem.	Designing a system to detect spam emails.
Analytical research	Research that analyzes existing facts, data, or information to draw conclusions.	Analyzing past cyberattack records to identify patterns.
Conceptual research	Research based on ideas, concepts, theories, and models rather than direct data collection.	Proposing a conceptual framework for ethical AI governance.
Empirical research	Research based on observation, experiment, measurement, or real data.	Testing a chatbot with 500 users and analyzing response accuracy.
Experimental research	Research where the researcher manipulates variables and observes effects.	Testing whether method A improves accuracy more than method B.

Type	Meaning	Simple example
Non-experimental research	Research where variables are observed without manipulation.	Surveying student opinions about e-learning.
Prospective research	Research that follows participants or data forward into the future.	Tracking students for one year to see how study habits affect performance.
Retrospective research	Research that examines past records or historical data.	Analyzing last five years of hospital patient records.
Exploratory research	Initial study when the problem is new or not well understood.	Interviewing users to understand why they abandon an app.
Descriptive research	Research that describes what exists, how often it occurs, or what characteristics it has.	Finding the percentage of students using AI tools for study.
Qualitative research	Research using non-numerical data such as interviews, observations, documents, and themes.	Studying student experiences through open-ended interviews.
Quantitative research	Research using numerical data and statistical analysis.	Measuring test scores before and after using an app.
Mixed method research	Research combining qualitative and quantitative methods.	Survey scores plus interviews to understand reasons behind the scores.
Survey research	Research that collects data from many respondents using questionnaires or forms.	Google Form on student internet usage.
Case study	Deep study of a single person, group, organization, project, system, or event.	Studying how one college adopted an LMS.
Action research	Research where practitioners study and improve their own practice.	A teacher tests a new classroom technique and improves it over cycles.
Evaluation research	Research that judges the effectiveness of a program, policy, tool, or intervention.	Evaluating whether a digital attendance system reduced proxy attendance.
Historical research	Research using past documents/events to understand development over time.	Studying the evolution of operating systems.

Common mistake
Students often confuse qualitative with theoretical. Qualitative research can still be empirical because interviews and observations are real data. Quantitative means numbers; qualitative means meanings, experiences, themes, or descriptions.

3.5 Research Process

Choose area -> Identify problem -> Review literature -> Define objectives -> Frame questions/hypothesis -> Design methodology -> Collect data -> Analyze data -> Interpret findings -> Report and publish

1. **Select a broad area:** Example: AI in education.
2. **Narrow it into a problem:** Example: Does AI feedback improve programming assignment performance?
3. **Review literature:** Read previous papers and identify what is already known.
4. **Find the research gap:** Identify what is missing, weak, outdated, or not studied in your context.
5. **Set objectives:** Write what the study will achieve.
6. **Decide methodology:** Choose survey, experiment, dataset analysis, case study, etc.
7. **Collect and analyze data:** Use suitable tools and statistics.
8. **Interpret results:** Explain what the results mean, not only what numbers are obtained.
9. **Report outcome:** Write findings, limitations, conclusion, and future scope.

3.6 Induction and Deduction in Research

Approach	Direction of thinking	Meaning	Example
Induction	Specific observations -> general theory	The researcher observes patterns and builds a general conclusion.	After interviewing many students, the researcher concludes that lack of feedback reduces motivation.
Deduction	General theory -> specific test	The researcher starts with theory or hypothesis and tests it with data.	Theory says feedback improves learning; researcher tests whether feedback improves programming scores.

> **Exam answer**
> Induction moves from observations to generalization, while deduction moves from theory or hypothesis to testing. In research, both are useful: induction helps generate ideas and deduction helps test them.

3.7 Introduction to Research Tools

Purpose	Tools / examples	Use in research
Finding literature	Google Scholar, Scopus, Web of Science, IEEE Xplore, ACM Digital Library, PubMed	Search research papers and identify state of the art.
Reference management	Zotero, Mendeley, EndNote	Store citations and create bibliography.
Writing	MS Word, Google Docs, LaTeX/Overleaf	Prepare research reports and manuscripts.
Data collection	Google Forms, Microsoft Forms, questionnaires, interview schedules, sensors, logs	Collect primary data or user responses.
Data analysis	Excel, SPSS, R, Python, Jupyter Notebook, RapidMiner, Orange	Clean data, compute statistics, build models, visualize results.
Plagiarism checking	Turnitin, Ouriginal, Grammarly plagiarism checker	Check similarity and originality.
Visualization	Excel charts, Python Matplotlib, Tableau, Power BI	Present findings clearly.

3.8 Qualities of a Good Researcher

- **Curiosity:** asks meaningful questions.
- **Critical thinking:** questions assumptions and checks evidence.
- **Honesty:** reports both positive and negative findings.
- **Patience and persistence:** continues even when results are not immediate.
- **Objectivity:** avoids bias and does not force data to match expectation.
- **Communication skill:** explains methods, results, and limitations clearly.
- **Ethical awareness:** protects participants, privacy, and intellectual property.
- **Technical competence:** uses appropriate tools and methods carefully.

Quick revision - Unit 1
Research must be systematic, logical, empirical, objective, ethical, reliable, valid, and useful. Know the differences among basic/applied, qualitative/quantitative, experimental/non-experimental, empirical/conceptual, exploratory/descriptive, prospective/retrospective, and mixed methods.

4. Unit 2 - Problem Identification and Literature Review

4.1 Research Problem

A research problem is a clear statement of an unanswered question, difficulty, gap, or issue that needs systematic investigation. It is the starting point of research. Without a clear problem, the study becomes directionless.

> **Exam definition**
> A research problem is a specific issue, difficulty, unanswered question, or knowledge gap that a researcher wants to investigate using systematic methods.

Good research problem	Poor research problem
"Does weekly AI-based feedback improve first-year programming performance?"	"AI and education"
"What factors affect adoption of digital payments among rural shopkeepers?"	"Digital payments are useful"
"How accurately can a CNN model classify traffic signs under low-light conditions?"	"Make a traffic sign project"

4.2 Features of a Good Research Problem

- **Clear and precise:** The reader should understand exactly what is being studied.
- **Researchable:** Data or evidence can be collected.
- **Feasible:** Possible within time, budget, skills, and access.
- **Significant:** Solving it adds value to knowledge or practice.
- **Original or contextually new:** It addresses a gap, new setting, new method, or new comparison.
- **Not too broad or too narrow:** It should be manageable.
- **Ethically acceptable:** It should not harm participants or misuse data.

4.3 Choosing an Appropriate Problem Area

1. Start with a broad area you understand or can access, such as AI in education, cybersecurity awareness, e-commerce, social media behavior, or data mining.
2. Read 5-10 recent papers to see common problems and methods.
3. Identify what is missing: a population not studied, a method not compared, a dataset limitation, lack of real-world evaluation, or contradictory findings.

4. Check feasibility: Can you get data? Can you analyze it? Is the topic too costly or sensitive?
5. Convert the topic into a focused question using population, variables, method, and context.

> **Formula for narrowing a topic**
> `Broad area + specific population/dataset + variable/method + expected analysis + context`
> Example: "AI chatbots" becomes "Impact of AI chatbot feedback on Python assignment completion among first-year BCA students."

4.4 Types of Research Problems

Type	Typical question	Example
Descriptive problem	What is happening?	What percentage of students use AI tools for exam preparation?
Relational problem	Is there a relationship?	Is screen time related to academic performance?
Difference problem	Is there a difference between groups?	Do online and offline learners differ in test scores?
Causal problem	Does X cause/change Y?	Does gamified practice improve coding accuracy?
Casuist/value problem	What is right or wrong in a moral situation?	Is it ethical to use student activity data without explicit consent?

4.5 Sources of Research Articles

Source	What it gives	Exam/viva point
Google Scholar	Broad search across scholarly papers, books, and citations.	Good starting point but results must be filtered for quality.
Scopus	Curated abstract and citation database.	Useful for high-quality indexed literature.
Web of Science	Citation database and journal indexing.	Useful for citation analysis and impact.
IEEE Xplore	Engineering, computer science, electronics papers.	Very useful for CS research.
ACM Digital Library		

Source	What it gives	Exam/viva point
	Computer science research papers and conference proceedings.	Useful for algorithms, HCI, systems, AI, software engineering.
SpringerLink / ScienceDirect	Journal articles and book chapters.	Useful for methodology and domain papers.
Institutional repositories	Theses, dissertations, reports.	Can help understand methodology and background.

4.6 Literature Review

A literature review is a systematic study of previous research related to your topic. It is not a simple summary of papers. It identifies what is already known, what methods have been used, where results agree or disagree, and what gap remains.

Step	What to do	Output
Search	Use keywords, synonyms, Boolean operators, and databases.	List of relevant papers.
Screen	Read title, abstract, keywords, year, journal/conference.	Remove irrelevant or low-quality papers.
Read deeply	Focus on introduction, methodology, results, limitations.	Notes on problem, method, dataset, findings.
Compare	Group papers by method, dataset, domain, result, or limitation.	Pattern table or literature matrix.
Identify gap	Find what is missing, weak, contradictory, or outdated.	Research gap and justification.

> **State of the art**
> State of the art means the current best or most advanced knowledge, methods, tools, datasets, or findings in a research area. In a paper, it is usually discussed in the literature review or related work section.

4.7 Research Gap

A research gap is the missing or insufficient part in existing literature that justifies the need for a new study. A gap may be theoretical, methodological, empirical, contextual, practical, or data-related.

Gap type	Meaning	Example
Population gap	A group has not been studied.	Most studies focus on university students, but not rural BCA students.
Context gap	A setting/geography/domain is missing.	AI feedback studied in US universities, not Indian colleges.
Method gap	Previous studies used limited or weak methods.	Only surveys were used; no experiment was conducted.
Data gap	Dataset is small, outdated, biased, or unavailable.	Model trained only on daytime images, not low-light images.
Contradictory findings	Earlier studies disagree.	Some studies say screen time harms scores; others find no effect.
Practical gap	Research exists but has not been tested in real use.	Algorithm works in lab but not deployed with real users.

4.8 Objectives, Research Questions, and Hypotheses

Element	Meaning	How to write	Example
Objective	What the study aims to achieve.	Start with action verbs: To identify, examine, compare, analyze, evaluate.	To evaluate the effect of AI feedback on Python assignment scores.
Research question	The question the study tries to answer.	Use what, how, why, to what extent, is there a relationship.	Does AI feedback improve Python assignment performance?
Hypothesis	A testable prediction about expected relationship/ difference.	State expected outcome; can be null or alternative.	H1: Students using AI feedback will score higher than students using normal feedback.

Difference to remember
Problem tells what issue exists. Objectives tell what the researcher will do. Research questions ask what the study wants to answer. Hypothesis predicts what the result may be.

Quick revision - Unit 2
A strong research problem is clear, researchable, feasible, significant, and ethically acceptable. Literature review helps identify the state of the art and research gap. Objectives and research questions must directly match the problem and methodology.

5. Unit 3 - Data Analytics and Statistics for Research

5.1 Data Analytics in Research

Data analytics is the process of exploring, organizing, cleaning, transforming, analyzing, and interpreting data so that useful findings can be drawn. In research methodology, data analytics connects raw data to valid conclusions.

5.2 Dataset Exploration and Organization

Task	Meaning	Example
Understand variables	Identify columns/features and their meaning.	Age, gender, test score, study hours.
Check data type	Numerical, categorical, text, date, image, audio, etc.	Age is numerical; gender is categorical.
Check size	Number of rows and columns.	500 responses and 12 variables.
Check missing values	Find blanks or unavailable entries.	Income column has 20 missing values.
Check outliers	Find unusual values.	Age = 250 is likely incorrect.
Check balance	Class distribution in dataset.	90% non-spam and 10% spam emails.
Summarize	Use statistics and plots.	Mean score, median score, frequency table.

5.3 Data Preprocessing

Data preprocessing prepares raw data for analysis. Raw data is often incomplete, noisy, inconsistent, duplicated, or stored in different formats. Without preprocessing, results may be misleading.

Raw data -> Cleaning -> Integration -> Transformation -> Reduction -> Analysis-ready data

Preprocessing step	Purpose	Common methods
Data cleaning	Remove or correct errors, missing values, noise, duplicates, inconsistencies.	Imputation, deletion, outlier detection, smoothing, validation.
Data integration	Combine data from multiple sources.	Schema matching, entity identification, resolving duplicate records.
Data transformation	Convert data into suitable form.	Normalization, standardization, encoding, aggregation.
Data reduction	Reduce data size while preserving important information.	Feature selection, dimensionality reduction, sampling, compression.
Data discretization	Convert continuous values into intervals/classes.	Age into child/teen/adult, marks into low/medium/high.

Handling missing data

Method	When useful	Risk
Remove record	When very few rows are missing and deletion will not bias data.	Can reduce sample size and bias result.
Fill manually	Small dataset and correct value can be verified.	Time-consuming.
Global constant	When missing itself is meaningful.	May create artificial category.
Mean/median/mode imputation	Simple numerical/categorical replacement.	Can reduce variance and hide real pattern.
Regression/model-based imputation	When missing value can be predicted from other variables.	Can add model bias.

Handling noisy data

Technique	Meaning	Example
Binning	Sort data and group into bins; smooth values by mean, median, or boundary.	Prices 4,8,15 can be replaced by bin mean 9.

Technique	Meaning	Example
Regression	Fit a line/curve and smooth values according to the model.	Correct a noisy temperature-humidity reading.
Clustering/outlier analysis	Detect points far from normal groups.	A student age of 200 in college data is an outlier.
Human inspection	Manually check suspicious values.	Confirm whether 0 marks means absent or actual zero.

5.4 Descriptive Statistics

Descriptive statistics summarize data. They do not prove cause, but help describe what the data looks like.

Measure	Meaning	Formula / method	Use
Mean	Average value.	Sum of values / number of values.	Average marks.
Median	Middle value after sorting.	Middle value or average of two middle values.	Useful when outliers exist.
Mode	Most frequent value.	Value appearing most often.	Most common category.
Range	Difference between maximum and minimum.	Max - Min.	Simple spread measure.
Variance	Average squared deviation from mean.	Measures spread around mean.	Higher variance means more variability.
Standard deviation	Square root of variance.	Typical distance from mean.	Common measure of variability.

5.5 Measures of Association

Measures of association show whether two variables are related. Association does not automatically mean causation.

Measure	Use	Interpretation
Correlation	Measures strength and direction of relationship between numerical variables.	Positive: both increase; negative: one increases, other decreases; near zero: weak linear relation.

Measure	Use	Interpretation
Chi-square association	Checks association between categorical variables.	Example: gender and preferred learning mode.
Regression coefficient	Shows how much outcome changes with predictor.	Study hours may predict marks.

5.6 Inferential Statistics

Inferential statistics uses sample data to make conclusions about a larger population. It includes estimation of population parameters and hypothesis testing.

Term	Meaning	Example
Population	Entire group about which conclusion is made.	All BCA students in India.
Sample	Subset selected from population.	200 BCA students from 5 colleges.
Parameter	Numerical value describing population.	True average study hours of all BCA students.
Statistic	Numerical value computed from sample.	Average study hours of 200 sampled students.
Confidence interval	Range likely to contain population parameter.	Mean score is estimated between 65 and 70.
p-value	Probability of observing results if null hypothesis is true.	Small p-value suggests evidence against H0.

5.7 Hypothesis Testing

1. State the null hypothesis H0, usually saying there is no difference/effect/relationship.
2. State the alternative hypothesis H1, saying there is a difference/effect/relationship.
3. Choose significance level, commonly 0.05.
4. Select a statistical test based on data type and research question.
5. Calculate test statistic and p-value.
6. Make decision: reject H0 if p-value is less than alpha; otherwise fail to reject H0.
7. Interpret result in research language, not only statistical language.

Research question	Possible test	Example
Difference between two group means	t-test	Online vs offline class scores.

Research question	Possible test	Example
Difference among more than two group means	ANOVA	Scores across three teaching methods.
Association between categorical variables	Chi-square test	Gender and learning preference.
Relationship between two numerical variables	Correlation/ regression	Study hours and marks.

Quick revision - Unit 3
Data analytics includes exploration, organization, preprocessing, analysis, and interpretation. Know data cleaning, integration, transformation, reduction, missing values, noisy data, descriptive statistics, association, inferential statistics, population/sample, parameter/statistic, and hypothesis testing.

6. Unit 4 - Presenting Research Outcomes and Research Paper Structure

6.1 Meaning of Presenting Research Outcomes

Presenting research outcomes means organizing and communicating the results of a study clearly so that readers can understand the problem, method, data, analysis, findings, limitations, and conclusion. A good presentation of outcomes is honest, structured, evidence-based, and easy to verify.

6.2 Essential Elements of a Research Paper

Part	Purpose	What to look for in exam
Title	Shows topic and focus of study.	Research area, key variables, population/context.
Abstract	Short summary of problem, method, results, conclusion.	Fastest place to identify methodology and findings.
Keywords	Main terms and research area.	Useful for identifying area and variables.
Introduction	Background, problem, importance, objectives.	Research problem, objectives, research questions.
Literature review / related work	Previous studies and gaps.	Research gap and state of the art.
Methodology	Design, sample, data, tools, procedure, analysis method.	Methodology, sampling, data collection, instruments.
Results	Outputs from analysis.	Findings, tables, graphs, model performance.
Discussion	Interpretation of results.	Meaning, comparison with previous studies, implications.
Limitations	Weaknesses or boundaries.	Sample size, dataset bias, limited scope, short duration.
Conclusion	Final answer to the research question.	Main takeaway and practical implication.
Future work	How study can be extended.	Improvements and next steps.

Part	Purpose	What to look for in exam
References	Sources cited.	Quality and recency of literature.

6.3 Data Analysis and Interpretation

Analysis means applying techniques to data, while interpretation means explaining what the results mean in relation to the research problem. For example, a model accuracy of 92% is an analysis result; saying that the model is promising but must be tested on larger datasets is interpretation.

Result type	How to present	Common mistake
Tables	Use clear headings, units, and labels.	Presenting table without explaining what it means.
Graphs	Use simple chart suitable for data.	Using fancy chart that hides meaning.
Statistical result	Report test, value, significance, interpretation.	Only writing p-value without research meaning.
Model performance	Report accuracy, precision, recall, F1, confusion matrix as needed.	Only reporting accuracy for imbalanced dataset.
Qualitative themes	Present themes with short evidence or quotes.	Listing opinions without grouping into themes.

6.4 Identifying Weaknesses of a Study

Weaknesses are not always failures. They are honest boundaries of the study. A good research paper should mention limitations so that readers know how far the findings can be generalized.

- Small sample size or narrow population.
- Short time duration.
- Use of self-reported data, which may contain bias.
- Limited dataset or dataset from one region/platform.
- No comparison with strong baseline methods.
- Only correlation found, not causation.
- Ethical or privacy constraints limiting data access.
- Lack of real-world deployment or external validation.

6.5 Drawing Conclusions

A conclusion should directly answer the research question. It should not introduce new results. It should summarize the main finding, explain significance, mention limitation briefly, and suggest future direction if needed.

> **Conclusion format**
> This study investigated [problem]. Using [method/data], it found that [main finding]. The result suggests [meaning/practical implication]. However, [main limitation]. Future studies can [improvement].

Quick revision - Unit 4
In an unseen paper, use the paper structure as a map. Abstract gives overview, introduction gives problem/objectives, related work gives gap, methodology gives design/sample/data/tools, results and discussion give findings, limitation/future work sections give weaknesses and improvements.

7. Unit 5 - Publication Process, Peer Review, Citation Analysis, and Plagiarism

7.1 Journal Submission and Review Process

Prepare manuscript -> Submit to journal -> Editor screening -> Peer review -> Decision -> Revision -> Acceptance / rejection -> Publication

Stage	What happens	Possible outcome
Submission	Author submits manuscript, cover letter, figures, declarations.	Manuscript enters editorial system.
Initial screening	Editor checks scope, quality, formatting, ethics, originality.	Desk reject or send to reviewers.
Peer review	Experts evaluate originality, methodology, validity, clarity, significance.	Accept, minor revision, major revision, reject.
Revision	Author responds to reviewer comments and improves paper.	Resubmission for editor/ reviewer check.
Final decision	Editor decides based on reviews and revisions.	Accepted or rejected.
Production	Copyediting, proofing, formatting, online publication.	Published article.

7.2 Peer Review

Peer review is the evaluation of a research manuscript by independent experts before publication. Its purpose is to improve quality, detect errors, judge originality, and maintain scholarly standards.

Type	Meaning	Advantage	Limitation
Single-blind review	Reviewer knows author identity, author does not know reviewer identity.	Reviewer can judge context and avoid direct pressure.	Reviewer bias may occur due to author identity or institution.

Type	Meaning	Advantage	Limitation
Double-blind review	Both author and reviewer identities are hidden from each other.	Reduces identity-based bias.	Sometimes identity can be guessed from topic or references.
Open review	Identities or review reports may be visible.	Transparency and accountability.	Reviewers may be less critical.

7.3 Journal Selection

- Match the journal scope and audience with the paper topic.
- Check indexing status such as Scopus, Web of Science, or field-specific databases.
- Check peer review policy, ethics policy, publication fees, and open access options.
- Check article structure and formatting requirements.
- Avoid predatory journals that promise unrealistic fast publication without proper review.
- Use journal finder tools from publishers when available.

7.4 Professional Research Societies

Professional research societies are organizations that support researchers in a field through conferences, journals, standards, workshops, networking, ethics guidelines, and professional development. In computer science, examples include ACM, IEEE Computer Society, and other domain-specific associations.

7.5 Scientometric Analysis, Citation Index, and Citation Analysis

Term	Meaning	Example / use
Scientometrics	Quantitative study of science, publications, citations, authors, institutions, and research impact.	Analyzing publication trends in AI ethics.
Citation	A reference from one scholarly work to another.	Paper A cites Paper B.
Citation index	Database tracking which papers cite which other papers.	Web of Science, Scopus, Google Scholar citation records.
Citation analysis	Studying citation patterns to assess influence, connection, or trend.	Finding most cited papers in cybersecurity education.

Term	Meaning	Example / use
Impact factor	Journal-level metric based on average citations to recent articles.	Used carefully, not as sole quality measure.
h-index	Author metric: h papers have at least h citations each.	h-index 10 means 10 papers each cited at least 10 times.

7.6 Plagiarism and Plagiarism Checkers

Plagiarism is the use of another person's words, ideas, data, figures, code, or structure without proper acknowledgement. It is an academic misconduct.

Type of plagiarism	Meaning	Example
Direct plagiarism	Copying exact text without citation or quotation.	Copy-paste paragraph from paper.
Mosaic plagiarism	Mixing copied phrases with own words without citation.	Changing a few words but keeping same structure.
Paraphrasing plagiarism	Rewriting someone's idea without citation.	Using the same argument from a paper but not citing it.
Self-plagiarism	Reusing own previous work without disclosure when not allowed.	Submitting same report in two courses.
Source-based plagiarism	Using wrong, fake, or incomplete citation.	Citing a source that was not actually used.
Data/code plagiarism	Using someone's dataset, code, or results without credit.	Copying GitHub code into project without acknowledgement.

Similarity index is not equal to plagiarism
A plagiarism checker such as Turnitin reports similarity. A high similarity may indicate plagiarism, but even low similarity can be unethical if ideas, data, or code are stolen without credit. For the practical report, aim for proper citation, paraphrasing, and similarity below the required threshold.

Quick revision - Unit 5
Know the journal review pipeline, peer review types, journal selection criteria, role of societies, scientometric terms, citation analysis, plagiarism types, and plagiarism checkers. In viva, explain why peer review and ethics improve research quality.

8. Unit 6 - Research Ethics

Research ethics refers to moral principles that guide how research is planned, conducted, analyzed, reported, and published. Ethics protects participants, researchers, institutions, society, and the credibility of knowledge.

Ethical issue	Meaning	Example in practical answer
Protection from harm	Research should not cause physical, psychological, social, legal, or financial harm.	Avoid collecting sensitive student data without safeguards.
Voluntary participation	Participants should not be forced to join the study.	Students should be free to refuse survey participation.
Informed consent	Participants should know purpose, risks, benefits, data use, and their rights.	Consent form before interview or questionnaire.
Right to privacy	Personal identity and private information should be protected.	Do not publish names, phone numbers, emails.
Confidentiality	Information should be accessed only by authorized people.	Store survey data securely and anonymize responses.
Conflict of interest	Personal, financial, or professional interest should not bias research.	Declare if funding company benefits from positive result.
Honesty with colleagues	Do not misrepresent data, contribution, authorship, or results.	Give co-authors proper credit.
Professional code of ethics	Follow discipline and institutional research guidelines.	Follow ACM/IEEE or university research standards.
Intellectual property rights	Respect ownership of ideas, data, code, images, and publications.	Cite papers and respect licenses.
Fraud and misconduct	Fabrication, falsification, plagiarism, image manipulation, fake authorship are misconduct.	Do not invent survey responses or edit graphs dishonestly.

How to identify ethics in an unseen paper

1. Check whether human participants, personal data, medical data, student data, social media data, or sensitive records are used.

2. Search for words such as consent, anonymized, approval, ethics committee, privacy, confidentiality, data protection, IRB, voluntary, permission.
3. If the paper uses public datasets, check whether privacy and licensing are mentioned.
4. If AI or data mining is involved, check bias, fairness, transparency, and misuse risk.
5. If no ethics statement is present, mention it carefully as a possible limitation, not as a confirmed violation.

Quick revision - Unit 6
Ethics is not only about human experiments. It also includes privacy, consent, data use, plagiarism, conflict of interest, intellectual property, honest reporting, and avoiding research fraud.

9. Practical Exam Toolkit - How to Analyze Any Research Paper

Use this chapter during practice. In the exam, do not read the paper like a novel from first page to last page. Read it strategically.

9.1 Research Paper Analysis Checklist

Item	Where to find it	Question to ask
Title and area	Title, keywords, abstract	What topic/domain is the paper about?
Research problem	Abstract, introduction, first 2-3 paragraphs	What issue or unanswered question is being addressed?
Objectives	End of introduction, section titled objectives/contribution	What does the study aim to achieve?
Research questions	Introduction, methodology, sometimes explicit RQ1/RQ2	What questions guide the study?
Hypothesis	Introduction/methodology/statistical analysis	Is there a testable prediction?
Literature review	Related work section	What previous studies are discussed?
Research gap	End of literature review/introduction	What is missing in previous work?
Methodology	Method/proposed work/experimental setup	What method/design is used?
Research design	Methodology section	Survey, experiment, case study, qualitative, quantitative, mixed, etc.
Population and sample	Participants/dataset section	Who/what was studied and how many?
Sampling technique	Methodology/data section	Random, convenience, purposive, stratified, etc.?
Data collection	Methodology	Survey, interview, observation, dataset, experiment, logs, documents.
Tools/instruments	Methodology/implementation	Questionnaire, interview guide, software, sensors, Python, SPSS, etc.
Data analysis	Analysis/results	Statistics, coding, thematic analysis, ML metrics, tests.
Results/findings	Results/discussion/abstract	What did the study find?
Limitations	Limitations/discussion/conclusion	What boundaries or weaknesses exist?

Item	Where to find it	Question to ask
Future scope	Conclusion/future work	How can it be improved or extended?
Ethics	Ethics statement/methodology	Consent, privacy, anonymization, approval, conflict of interest.
Conclusion	Conclusion section	What final answer does the paper give?
Practical implication	Discussion/conclusion	How can findings be used in real life?

9.2 Step-by-Step Paper Analysis Method

1. **Read title and keywords:** Identify area, variables, method clues, and population/context.
2. **Read abstract slowly:** Underline problem, method, data/sample, main result, conclusion.
3. **Read last paragraph of introduction:** Usually contains aim, objectives, contribution, or paper organization.
4. **Scan related work:** Do not read every line. Identify what previous studies did and what gap remains.
5. **Read methodology carefully:** This is the most important practical section. Identify research design, sample, data collection, tools, and analysis.
6. **Study results tables/figures:** Find main findings; do not waste time on every number.
7. **Read discussion/conclusion:** Identify meaning, implication, limitation, and future work.
8. **Fill the checklist:** Use short phrases first, then convert into full answer format.

9.3 Evidence Words That Reveal the Methodology

Words in paper	Likely methodology/design
questionnaire, respondents, Likert scale, Google Form	Survey research; usually quantitative or mixed.
interviews, focus group, themes, coding	Qualitative research.
control group, treatment group, pre-test, post-test	Experimental or quasi-experimental research.
dataset, features, model, training, testing, accuracy	Empirical computational/ML research.
case organization, one school/company/project	Case study.
existing records, past data, archives	Retrospective or historical/document analysis.
observation, field notes, behavior recorded	Observational research.
combines survey and interview	Mixed method research.

10. How to Read a Research Paper in 15-20 Minutes

Minute	Action	What to underline
0-2	Read title, keywords, abstract.	Research area, method words, sample/dataset, main finding.
2-5	Read introduction first and last paragraphs.	Problem, importance, objectives, contribution.
5-8	Scan related work/literature review.	Gap words: however, limited, few studies, not addressed, lack of.
8-13	Read methodology carefully.	Design, sample, population, data collection, tools, analysis techniques.
13-17	Read results tables/figures and discussion.	Main findings, statistical results, model metrics, themes.
17-20	Read conclusion, limitations, future work, ethics.	Limitations, implications, future research, privacy/ consent.

What to underline

- Aim/objective sentences: "This study aims to..."
- Gap sentences: "However, previous studies have not..."
- Method sentences: "We conducted a survey/experiment/interview..."
- Sample/data sentences: "Data were collected from..." or "The dataset contains..."
- Analysis sentences: "We used t-test/regression/thematic analysis/CNN..."
- Finding sentences: "Results show that..."
- Limitation sentences: "This study is limited by..."
- Future work sentences: "Future research should..."
- Ethics sentences: "Informed consent was obtained..." or "Data were anonymized..."

What to ignore when time is limited

- Long historical background unless it defines the problem.
- Every single citation in the literature review.
- Complex mathematical derivation unless the question asks for it.
- Appendix details unless sample/tool/questionnaire is there.
- Decorative figures that do not explain method or results.

Emergency answer trick
When unsure, write a cautious answer: "The paper appears to use [method] because [evidence from paper]." Avoid overclaiming. For example: "It appears to use convenience sampling because participants were selected from easily accessible college students, and no random selection procedure is mentioned."

11. Example Bank for Practical Questions

Use these examples to learn the pattern of answers. In the exam, replace the topic with the topic of the given paper.

10 Examples of Research Problems

Area	Example
AI in education	Many first-year programming students struggle to understand compiler errors, and it is unclear whether AI-generated feedback improves their learning outcomes.
Cybersecurity	College students use weak passwords, but the factors influencing their password behavior are not clearly understood.
E-commerce	Small local shops are adopting digital payments, but many face difficulties related to trust, cost, and technical knowledge.
Healthcare AI	Existing skin disease detection models may perform poorly on images captured under low-light mobile conditions.
Social media	Excessive short-video consumption may affect student attention span, but evidence in local college contexts is limited.
Cloud computing	Small businesses want cloud adoption, but they lack clarity about cost, security, and migration challenges.
Data mining	Customer churn prediction models often ignore behavioral changes over time.
Online learning	Online classes provide flexibility, but student engagement may decline without interactive tools.
Public services	Citizens face delays in accessing e-governance services, but the usability problems are not well documented.
Software engineering	Beginner programmers commit repeated syntax errors, but existing teaching tools do not provide personalized correction.

10 Examples of Research Objectives

No.	Example
1	To identify the common types of compiler errors made by first-year programming students.
2	To analyze the relationship between AI feedback usage and programming assignment performance.
3	To compare student engagement in online and offline classes.
4	To evaluate the accuracy of a machine learning model on a selected dataset.
5	To examine the factors influencing digital payment adoption among small shopkeepers.
6	To study the privacy concerns of students while using educational apps.

No.	Example
7	To investigate the effect of screen time on self-reported concentration levels.
8	To assess the effectiveness of a cloud-based file sharing system in a college department.
9	To explore the challenges faced by users while using e-governance portals.
10	To recommend improvements based on the findings of the study.

10 Examples of Hypotheses

No.	Example
1	H0: There is no significant difference in test scores between students using AI feedback and students using normal feedback.
2	H1: Students using AI feedback achieve significantly higher test scores than students using normal feedback.
3	H0: There is no relationship between daily screen time and academic performance.
4	H1: Higher daily screen time is negatively associated with academic performance.
5	H0: Gender has no significant association with preferred learning mode.
6	H1: Gender is significantly associated with preferred learning mode.
7	H0: Digital payment adoption is not influenced by perceived ease of use.
8	H1: Perceived ease of use positively influences digital payment adoption.
9	H0: The proposed model does not improve classification accuracy over the baseline model.
10	H1: The proposed model improves classification accuracy over the baseline model.

10 Examples of Limitations

No.	Example
1	The sample size is small, so the findings may not represent the entire population.
2	The study was conducted in one college only, limiting generalizability.
3	The data were self-reported, so responses may contain social desirability bias.
4	The study used a cross-sectional design, so causal conclusions cannot be made.
5	The dataset may contain class imbalance, affecting model evaluation.
6	The experiment duration was short and may not capture long-term effects.
7	The study did not compare the method with enough state-of-the-art baselines.
8	The research used convenience sampling, which may introduce selection bias.
9	Only quantitative data were collected, so deeper reasons behind responses are not known.

No.	Example
10	Privacy restrictions limited access to more detailed user data.

10 Examples of Methodologies

No.	Example
1	Survey methodology using structured questionnaire to collect numerical responses from students.
2	Experimental methodology using control and experimental groups to compare learning outcomes.
3	Case study methodology focusing on one institution or organization in depth.
4	Qualitative interview methodology to understand participant experiences and opinions.
5	Mixed method methodology combining questionnaire data with follow-up interviews.
6	Empirical machine learning methodology using dataset training, testing, and evaluation metrics.
7	Document analysis methodology using existing reports, policies, or published literature.
8	Observational methodology where user behavior is recorded without manipulation.
9	Retrospective methodology using past records or archival data.
10	Evaluation methodology assessing the effectiveness of a tool, program, or intervention.

10 Examples of Research Gaps

No.	Example
1	Previous studies focused on urban students, while rural college students remain underexplored.
2	Existing models were tested on clean datasets but not on noisy real-world data.
3	Most studies used surveys only and did not include interviews for deeper understanding.
4	There is limited research comparing AI feedback with teacher feedback in beginner programming.
5	Earlier studies reported accuracy but ignored fairness and bias.
6	Few papers examine long-term effects; most studies are short-term.
7	Existing literature lacks evidence from Indian higher education contexts.
8	Previous research focused on adoption but not on user trust and privacy concerns.
9	There is limited work using mixed methods to study the same problem.
10	The state-of-the-art methods have not been evaluated on low-resource devices.

10 Examples of Sampling Methods

No.	Example
1	Simple random sampling: each member has equal chance of selection. Example: randomly selecting 100 students from enrollment list.
2	Stratified sampling: population divided into strata, sample selected from each. Example: selecting students from each year.
3	Cluster sampling: groups/clusters are selected instead of individuals. Example: selecting 5 classes and surveying all students in them.
4	Systematic sampling: every kth member is selected. Example: every 10th student from a list.
5	Convenience sampling: easily available participants are selected. Example: students from researcher's own class.
6	Purposive sampling: participants selected because they have specific knowledge. Example: interviewing teachers using LMS.
7	Snowball sampling: participants refer other participants. Example: locating cybersecurity professionals through referrals.
8	Quota sampling: fixed number selected from categories. Example: 50 male and 50 female respondents.
9	Volunteer sampling: participants choose to join. Example: open online survey link.
10	Census: entire population is studied. Example: all students in a small department.

10 Examples of Data Collection Methods

No.	Example
1	Questionnaire: structured set of questions used for surveys.
2	Interview: direct conversation to collect detailed responses.
3	Observation: recording behavior or events as they occur.
4	Experiment: collecting data after applying treatment/intervention.
5	Existing dataset: using public or institutional datasets.
6	Document analysis: analyzing reports, policies, papers, logs, or records.
7	Focus group: group discussion guided by researcher.
8	Sensor/log data: automatic collection from devices, apps, or systems.
9	Test/assessment: collecting scores from exams or tasks.
10	Case records: using historical records from an organization.

10 Examples of Ethical Issues

No.	Example
1	Collecting student data without informed consent.
2	Publishing participant names or identifiable information.
3	Using private social media data without permission.
4	Forcing students to participate because teacher is conducting research.
5	Not disclosing conflict of interest with a sponsoring company.
6	Copying text, code, or figures without citation.
7	Fabricating survey responses to increase sample size.
8	Manipulating graphs to exaggerate results.
9	Using biased AI models that unfairly affect a group.
10	Storing sensitive data without password protection or anonymization.

10 Examples of Viva-style Short Answers

No.	Example
1	Q: What is research? A: A systematic investigation to answer questions or solve problems using evidence.
2	Q: What is research gap? A: A missing or insufficient area in existing literature that justifies a new study.
3	Q: What is methodology? A: The overall plan and methods used to conduct the research.
4	Q: What is sample? A: A subset of the population selected for study.
5	Q: What is hypothesis? A: A testable prediction about relationship, difference, or effect.
6	Q: What is literature review? A: A critical study of existing research related to the topic.
7	Q: What is peer review? A: Evaluation of a manuscript by subject experts before publication.
8	Q: What is plagiarism? A: Using someone else's work or ideas without proper credit.
9	Q: What is informed consent? A: Participant's voluntary agreement after understanding study purpose and risks.
10	Q: What is limitation? A: A weakness or boundary that affects interpretation or generalization of findings.

12. Ready-to-Use Answer Templates

Identify the methodology used in this paper.

Ready answer format
The paper uses a [qualitative/quantitative/mixed/experimental/survey/case study/empirical] methodology. This can be identified from the [methodology/experimental setup/data collection] section, where the authors mention [evidence: questionnaire/interview/dataset/experiment/model]. The method is suitable because [reason linked to objective].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

What are the limitations of this study?

Ready answer format
The main limitations of the study are [limitation 1], [limitation 2], and [limitation 3]. These limitations affect [generalizability/validity/accuracy/interpretation] because [explanation]. Future work can address this by [improvement].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

What is the research gap?

Ready answer format
The research gap is that previous studies have [what they did], but they have not adequately addressed [missing aspect]. The authors justify the present study by focusing on [new method/context/population/dataset/comparison].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

What are the objectives of the research?

> **Ready answer format**
> The objectives of the study are to [objective 1], [objective 2], and [objective 3]. These objectives are connected to the research problem because they help investigate [problem area] and produce evidence about [expected outcome].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

Which sampling technique is used?

> **Ready answer format**
> The study appears to use [sampling technique]. This is because the sample was selected by [evidence from paper]. If the paper does not clearly mention the technique, it can be inferred as [likely technique] based on [reason], but the lack of explicit sampling details is a limitation.

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

What is the data collection method?

> **Ready answer format**
> The data collection method used in the paper is [questionnaire/interview/observation/experiment/public dataset/logs/documents]. Evidence for this is [specific phrase/section]. This method is appropriate because [why it fits the objective].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

What are the findings of the paper?

> **Ready answer format**
> The major findings are [finding 1], [finding 2], and [finding 3]. These findings answer the research question by showing that [interpretation]. The most important result is [main result] because [significance].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

How can this study be improved?

> **Ready answer format**
> The study can be improved by increasing the sample size, using a more diverse population/dataset, adding stronger comparison methods, applying better statistical analysis, addressing ethical/privacy concerns, and validating results in real-world conditions. The best improvement for this paper specifically is [paper-specific improvement].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

Identify ethical considerations.

> **Ready answer format**
> The ethical considerations include [consent/privacy/confidentiality/conflict of interest/data security/IP rights]. The paper handles ethics by [evidence], or if not mentioned, the study should have included [ethical safeguard].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

Write conclusion of the paper.

> **Ready answer format**
> The paper concludes that [main conclusion]. This conclusion is based on [method/results]. It contributes to [field/practice] by [contribution], but the conclusion should be interpreted within [limitations].

How to reach the answer

1. Find the relevant paper section: abstract, introduction, methodology, results, discussion, conclusion, or limitations.
2. Underline evidence words and data points.
3. Convert evidence into research-methodology vocabulary.
4. Write a balanced answer: identify, justify, explain importance, and mention limitation if needed.

13. Viva Questions and Answers

Viva 1. What is research?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Research is a systematic investigation to discover knowledge or solve a problem.	Research follows a planned process: identifying a problem, reviewing literature, collecting and analyzing data, and drawing conclusions. It is based on evidence, not guesswork.	Speak slowly. Start with "systematic investigation" and add "evidence-based".	systematic, logical, evidence, problem, conclusion	Do not say research means only searching information online.

Viva 2. Why is literature review important?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It helps understand existing work and identify the research gap.	A literature review shows what has already been studied, what methods were used, what results were found, and what is still missing. It prevents duplication and justifies the new study.	Mention both existing knowledge and gap.	previous studies, gap, state of the art, justification	Do not call it a list of summaries only.

Viva 3. What is a research problem?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is a specific issue or unanswered question that needs investigation.	A research problem defines what the researcher wants to study. It should be clear, feasible, researchable, significant, and ethically acceptable.	Give a short example if asked.	unanswered question, difficulty, researchable, feasible	Do not make it too broad like "AI" or "education".

Viva 4. What is research gap?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is the missing or insufficient area in existing literature.	A research gap may occur when a population, method, dataset, context, comparison, or long-term effect has not been adequately studied.	Use the phrase "previous studies have..., however...".	missing, insufficient, previous studies, context, method	Do not confuse gap with limitation; gap exists before the new study.

Viva 5. What is methodology?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Methodology is the overall plan used to conduct the research.	It includes research design, sample, sampling technique, data collection method, tools, procedure, and data analysis technique.	List the components in order.	design, sample, data collection, tools, analysis	Do not say methodology means only software used.

Viva 6. Difference between method and methodology?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Method is a specific technique; methodology is the overall research plan.	For example, questionnaire is a method of data collection, while survey methodology includes research design, sample, questionnaire, procedure, and statistical analysis.	Use example; examiners like this.	specific technique, overall plan	Do not use both words as exactly same.

Viva 7. What is hypothesis?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
A hypothesis is a testable prediction about a relationship, difference, or effect.	In quantitative research, hypotheses are tested using data and statistical methods. H0 usually says no effect/ difference, while H1 says an effect/ difference exists.	Mention H0 and H1 if possible.	testable prediction, H0, H1, variable	Do not write a hypothesis as a vague opinion.

Viva 8. What is sample and population?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Population is the whole group; sample is the selected subset.	If the study is about all college students, that is the population. If 200 students are selected for data collection, those 200 are the sample.	Use a student example.	population, sample, subset, generalization	Do not say sample and population are same.

Viva 9. What is sampling?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Sampling is the process of selecting a subset from the population.	Sampling is needed when studying the whole population is not possible. It can be random, stratified, cluster, convenience, purposive, snowball, etc.	Name 2-3 types confidently.	selection, subset, random, convenience, purposive	Do not claim random sampling unless paper says random selection.

Viva 10. What is data collection method?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is the way data is gathered for the study.	Common methods include questionnaire, interview, observation, experiment, existing dataset, document analysis, logs, and tests.	Connect method to paper evidence.	questionnaire, interview, dataset, observation	Do not confuse data collection with data analysis.

Viva 11. What is data preprocessing?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is preparing raw data for analysis.	It includes data cleaning, integration, transformation, reduction, and discretization. It handles missing values, noise, outliers, duplicates, and inconsistent formats.	Use raw data -> clean data example.	cleaning, missing values, noisy data, transformation	Do not skip preprocessing in dataset-based studies.

Viva 12. What are descriptive statistics?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
They summarize data.	Mean, median, mode, range, variance, and standard deviation describe central tendency and variability.	Mention that they do not prove causation.	mean, median, mode, standard deviation	Do not treat mean as always best when outliers exist.

Viva 13. What is inferential statistics?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It uses sample data to make conclusions about a population.	It includes estimation, confidence intervals, hypothesis testing, p-values, and statistical tests such as t-test, chi-square, ANOVA, and regression.	Mention sample to population.	sample, population, hypothesis testing, confidence interval	Do not say it only describes the sample.

Viva 14. What is peer review?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is expert evaluation of a research paper before publication.	Peer reviewers check originality, methodology, validity, clarity, contribution, ethics, and significance. Their feedback helps the editor decide whether to accept, revise, or reject the paper.	Say it maintains quality.	reviewers, quality, validity, originality, publication	Do not say reviewers publish the paper directly.

Viva 15. Single blind vs double blind review?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
In single blind, reviewer knows author but author does not know reviewer. In double blind, both identities are hidden.	Double blind review tries to reduce identity bias. Single blind review is common but may be affected by author/institution bias.	Answer in comparison form.	identity, reviewer, author, bias	Do not reverse the meanings.

Viva 16. What is plagiarism?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Plagiarism is using someone else's words, ideas, data, code, or work without proper credit.	It includes direct copying, paraphrasing without citation, self-plagiarism, data plagiarism, code plagiarism, and source-based plagiarism.	Say it is misconduct.	copying, citation, originality, misconduct	Do not say changing words removes plagiarism.

Viva 17. What is informed consent?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is voluntary agreement after understanding the study.	Participants must know the purpose, procedure, risks, benefits, data use, confidentiality, and right to withdraw before agreeing.	Mention voluntary.	voluntary, purpose, risks, right to withdraw	Do not say consent is only a signature.

Viva 18. What is conflict of interest?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is a situation where personal or financial interest may bias research.	For example, if a company funds a study on its own product, the relationship must be declared so readers can judge possible bias.	Give company example.	bias, disclosure, funding, transparency	Do not say conflict always means fraud; it may be managed by disclosure.

Viva 19. What is citation analysis?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
It is the study of citation patterns to assess influence or trends.	Citation analysis examines how often and where papers are cited. It helps identify influential works, research trends, and relationships among studies.	Mention it should not be the only quality measure.	citations, influence, trends, impact	Do not judge paper quality only by citations.

Viva 20. How do you identify limitations in a paper?

Direct answer	Slightly detailed answer	How to speak confidently	Keywords	Mistake to avoid
Read limitation, discussion, conclusion, methodology, and future work sections.	Look for weaknesses such as small sample, limited context, self-reported data, short duration, dataset bias, lack of baseline comparison, or inability to prove causation.	Use phrases: "The study is limited by..."	sample size, bias, scope, generalization	Do not invent limitations unrelated to the paper.

14. Final Revision Sheet

One-page concept map

Research area -> Research problem -> Literature review ->
Research gap -> Objectives/RQs/Hypothesis -> Methodology ->
Data analysis -> Findings -> Limitations -> Conclusion/Future work

Must-know differences

Pair	Difference
Research problem vs research gap	Problem is the issue being studied; gap is what is missing in previous research.
Objective vs research question	Objective states what the study will do; research question asks what the study will answer.
Method vs methodology	Method is a specific technique; methodology is the complete plan.
Population vs sample	Population is the whole group; sample is the selected subset.
Data collection vs data analysis	Collection gathers data; analysis interprets it.
Qualitative vs quantitative	Qualitative uses words/themes; quantitative uses numbers/statistics.
Descriptive vs inferential statistics	Descriptive summarizes sample data; inferential generalizes from sample to population.
Limitation vs future scope	Limitation is current weakness; future scope is how to improve or extend.
Plagiarism vs citation	Plagiarism is using work without credit; citation gives proper credit.
Single blind vs double blind	Single blind hides reviewer from author; double blind hides both reviewer and author identities.

Last-minute answer skeleton

Universal practical answer
According to the paper, [answer]. This is evident from [section/evidence]. It is important because [reason]. However, [limitation/caution if any]. Therefore, [short conclusion].

Final checklist before entering practical exam

- Can I define research, methodology, literature review, research gap, hypothesis, sample, population, plagiarism, peer review, ethics?
- Can I identify methodology from evidence words like survey, interview, dataset, experiment, model, questionnaire?
- Can I write objectives using action verbs?
- Can I write limitations without blaming the paper unfairly?
- Can I explain data preprocessing and basic statistics?
- Can I distinguish result, finding, discussion, conclusion, and future work?
- Can I speak direct answers first, then detailed explanation?

Confidence line for viva
When you do not know the exact term, say: "Based on the evidence in the paper, it appears to be..." This sounds careful and academic. Do not guess loudly; justify calmly.